

Class 17: Genome informatics

Ami Pant (A15991358)

Identifying Genetic Variants of Interest

Q1: What are those 4 candidate SNPs?

The four candidate SNPs are rs12936231, rs8067378, rs9303277, and rs3816470.

Q2: What three genes do these variants overlap or effect?

The rs12936231 variant overlaps with ZPBP2, rs9303277 overlaps with IKZF3, and rs3816470 also overlaps with IKZF3. The rs8067378 variant does not overlap with a specific gene, but it falls between the ZPBP2 and GSDMB genes.

Q3: What is the location of rs8067378 and what are the different alleles for rs8067378?

rs8067378 is located on chromosome 17 at position 39895095 and its different alleles are A, C, and G.

Q4: Name at least 3 downstream genes for rs8067378?

GSDMB, ORMDL3, and LRRC3C are genes downstream of rs8067378.

Q5: What proportion of the Mexican Ancestry in Los Angeles sample population (MXL) are homozygous for the asthma associated SNP (G|G)?

0.1384615 of the Mexican Ancestry in Los Angeles sample population are homozygous for the asthma associated SNP (G|G).

Q6: Back on the ENSEMBLE page, use the “search for a sample” field above to find the particular sample HG00109. This is a male from the GBR population group. What is the genotype for this sample?

This sample is homozygous for the asthma associated SNP (G|G).

Initial RNA-Seq Analysis

Q7: How many sequences are there in the first file? What is the file size and format of the data? Make sure the format is fastqsanger here!

There are 3863 sequences in the first file. This is a fastqsanger file of size 741.9 KB

Q8: What is the GC content and sequence length of the second fastq file?

The GC content is 54% and the sequence length is 50-75 in the second fastq file.

Q9: How about per base sequence quality? Does any base have a mean quality score below 20?

None of the bases have a mean quality score below 20. None of the median scores are below the blue line, so the data set does not need to be trimmed.

Mapping RNA-Seq reads to genome

Q10: Where are most the accepted hits located?

The most accepted hits map to GSDMP, ORMDL3, ZPBP2, and PSMD3.

Q11: Following Q10, is there any interesting gene around that area?

In this area, ZPBP2, GSDMB, GSDMA, and ORMDL3 genes show up which are genes at which some SNPs overlap with or are near.

Q12: Cufflinks again produces multiple output files that you can inspect from your right-hand side galaxy history. From the “gene expression” output, what is the FPKM for the ORMDL3 gene? What are the other genes with above zero FPKM values?

The FPKM for ORMDL3 is 136853. The other genes with FPKM values above zero are ZPBP2, GSDMA, GSDMB, and PSMD3.